THORAXSCAN · STUDENT GUIDE

Reading attention maps

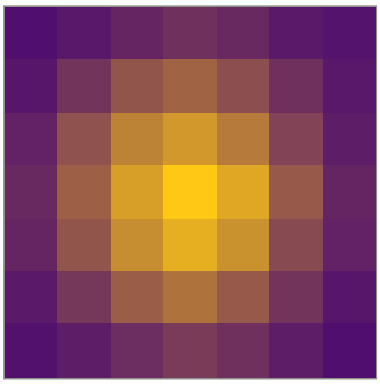
What ThoraxScan's attention heatmap shows, where you will find it, and how to use it responsibly while learning chest X-ray interpretation.



Purple: lower relative attention — Yellow: higher relative attention

> **Key idea:** the attention map shows where the model's internal attention was concentrated across the image. **It doesn't show where disease is.** It is a learning aid, not a diagnosis, and it never replaces your own image review.

1. What the map is



1. **7×7 grid.** The Vision Transformer splits the X-ray into a 7×7 grid of patches, so every map has 49 values.
2. **Four-layer attention rollout.** Attention from four of the model's layers is combined and averaged over queries into one map.
3. **Normalized per image.** The 49 values are scaled from 0 to 1 within that single image. Warmer (yellow) means higher *relative* attention.

What the colors do and do not mean

On the map	Meaning
Yellow / brighter patches	The model attended more to these patches than to others *in this image*.
Purple / darker patches	Less relative attention. This does **not** mean the area is normal or irrelevant.
Comparing two images	Not meaningful. Each map is scaled on its own, so a yellow patch in one image is not "stronger" than one in another.

> **Global and class-agnostic.** ThoraxScan produces one map per image. It is **not** specific to any predicted finding, so it cannot tell you where a particular finding is or why the model predicted it. Attention values are not probabilities and not a measure of model confidence.

2. Where you will see it

Attention Maps page

Opened from the main menu. It explains the method and shows a sample chest X-ray with an overlay labelled "*Illustrative example – not model output*". The sample uses hand-picked values to demonstrate the display. No model was run on it.

Learning Mode (ungraded practice)

1. Choose or drop a local chest X-ray (JPEG or PNG, up to 6 MiB).
2. Select **Analyze**. The model runs in your browser, and your image stays in browser memory.
3. The analysis result shows the findings and a **Model attention** overlay together.

Tip: write down your own observations *before* you select Analyze, so you can compare them honestly afterwards.

Assigned activities

1. Review the case X-ray and submit your final evaluation.
2. AI feedback is **locked until you submit**. After submission it shows "Analysis pending" until the analysis runs and is released.
3. Once released, the **AI feedback** panel shows the findings and the **Model attention** overlay. If analysis fails, select **Retry analysis**.
4. Your professor's evaluation, grade and comments remain the assessment. Use the map for reflection.

Professor submission review

Professors see a **Relative global attention** overlay next to the case X-ray while reviewing student submissions. They can use it to discuss agreement and disagreement with student observations.

Overlay controls

Control	Use it to
Show attention overlay	Toggle the heatmap off to see the original X-ray unobstructed.
Overlay opacity (default 55%)	Fade the colors so the anatomy underneath stays visible.
Method and 7×7 data	Expand to read the exact 49 normalized values, by row and column.

> **No map shown?** If you see "*This model release does not export attention data,*" the active model version does not produce attention maps. The prediction results are still available.

3. How to read it

1. Observe
Describe what is visible on the X-ray yourself.

2. Locate
State where the overlay is warmest, using anatomical location words.

3. Compare
Check the model's focus against your own observations.

4. Qualify
Note agreement or mismatch, and don't treat attention as evidence.

Do	Avoid
Use the map as one visual cue alongside the original image and your course knowledge.	Calling a yellow patch a confirmed abnormality.
Describe location carefully, e.g. "attention is concentrated over the lower right lung field."	Assuming purple areas are disease-free.
Treat a mismatch as a reason to re-check the image.	Reading the map as the reason for a specific predicted finding.
Keep reviewing the whole image, including areas with low attention.	Comparing color intensity between different images.

Discussion prompts

- Which anatomy or region received the most relative attention?
- Does that region correspond to something you can see on the X-ray?
- What did you observe outside the highlighted area?
- Where does your interpretation differ from the model's focus, and what should you re-check?

4. Use limits

ThoraxScan is an assistive learning tool for Radiologic Technology students. Its AI output, including attention maps, is for educational use only. It is not a diagnosis and must not guide clinical decisions or patient care. Follow your instructor's directions and course-approved interpretation methods.

> **Quick reference.** 7×7 grid, 49 values · four-layer attention rollout · normalized per image · yellow = higher, purple = lower *relative* attention · global and class-agnostic · not a probability · not a disease location.